Electricity Bill Minimization via Linear Programming

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Outline

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Motivation

- Electricity cost depends on **when** energy is used, not just total consumption.
- Peak hours: high tariff (high grid demand).
- Off-peak hours: low tariff (low demand).
- **Goal**: Shift flexible appliance usage to cheaper periods without reducing service quality.
- Approach: Linear programming formulation and solution in R using `lpSolve`.

Project Overview

- Data: Monthly consumption of 6 appliances (fan, refrigerator, AC, TV, monitor, motor pump).
- Tariff structure: peak rate $= 1.2\bar{c}$, off-peak rate $= 0.8\bar{c}$.
- Key finding: **33.3% cost reduction** (annual saving of Rs. 7,958,880).
- Full reproducibility: Kaggle dataset, GitHub repository, R script.

Appliance Demand and Tariff Structure

Appliance	Requirement (kWh)	Nature
Fan	364	Flexible
Refrigerator	564	Continuous
Air Conditioner	475	Partially flexible
Television	253	Flexible
Monitor	253	Flexible
Motor Pump	364	Scheduleable

Table 1: Representative annual demand and flexibility

- Peak tariff: $c^{(p)} = 1.2\bar{c}$ Off-peak tariff: $c^{(o)} = 0.8\bar{c}$

Tariff Construction Details

The average tariff $\bar{c}$ is computed from the dataset. Two effective rates are derived:

$$c^{(p)} = 1.2\,\bar{c}, \qquad c^{(o)} = 0.8\,\bar{c}$$

This simple gap creates a clear economic incentive to shift load.

- The ratio $c^{(o)}/c^{(p)} = 2/3$ directly determines the maximum possible saving.
- In reality, time-of-use tariffs vary by hour; here we use a stylised peak/off-peak split.

Linear Programming Model

Decision Variables

x_i = energy (kWh) allocated to the cheaper off-peak schedule for appliance i

Objective

Minimize total cost:

$$\min Z = \sum_{i=1}^{n} c_i^{(o)} x_i$$

Constraints

Demand must be satisfied:

$$x_i \geq R_i \quad \forall i, \qquad x_i \geq 0$$

Optimal Solution

Because all objective coefficients are positive, the optimum is attained at the lower bounds:

$$x_i^* = R_i \quad \text{for all } i$$

In matrix notation:

$$\min \mathbf{c}^\top \mathbf{x} \quad \text{s.t.} \quad I_n \mathbf{x} \geq \mathbf{R},\ \mathbf{x} \geq \mathbf{0}$$

The constraint matrix is diagonal, so each appliance is treated independently.

Cost Comparison Formulas

Baseline Cost (all peak)

$$Z_{\text{before}} = \sum_{i=1}^{n} c_i^{(p)} R_i$$

Optimized Cost (all off-peak)

$$Z_{\text{after}} = \sum_{i=1}^{n} c_i^{(o)} R_i$$

Savings

$$S = Z_{\text{before}} - Z_{\text{after}}$$

$$\text{Saving\%} = \frac{S}{Z_{\text{before}}} \times 100$$

Illustrative Calculation (Fan Example)

Suppose a fan consumes $R = 3$ kWh, with $c^{(p)} = 6$ Rs./kWh and $c^{(o)} = 4$ Rs./kWh.

$$Z_{\text{before, fan}} = 6 \times 3 = 18 \quad \text{Rs.}$$

$$Z_{\text{after, fan}} = 4 \times 3 = 12 \quad \text{Rs.}$$

Saving for this appliance: $18 - 12 = 6$ Rs. The same linear principle applies to all appliances together.

Why Exactly 33.3%?

With $c^{(p)} = 1.2\bar{c}$ and $c^{(o)} = 0.8\bar{c}$:

$$\frac{Z_{\text{after}}}{Z_{\text{before}}} = \frac{0.8\bar{c}}{1.2\bar{c}} = \frac{2}{3}$$

Thus,

$$\text{Saving\%} = 1 - \frac{2}{3} = \frac{1}{3} \approx 33.3\%$$

The savings are completely determined by the tariff gap, not by demand reduction.

Computational Workflow

1. **Data preparation**: Load monthly appliance data, convert to numeric, compute annual demand R_i using column sums.
2. **Tariff construction**: Derive peak and off-peak rates from average tariff $\bar{c}$.
3. **Optimization**: Solve LP with `lp()` from `lpSolve` using a diagonal constraint matrix.
4. **Post-processing**: Calculate costs, savings, efficiency ratio $E = S/\sum R_i$, and generate comparative plots.

All steps are fully reproducible via the provided R script.

Key Numerical Outcomes

Metric	Value
Baseline cost (all peak)	Rs. 23,876,640
Optimized cost (all off-peak)	Rs. 15,917,760
Total savings	Rs. 7,958,880
Percentage savings	33.3%
Efficiency ratio	3.38 Rs./kWh

Table 2: Summary of optimization results

- Savings are achieved purely by scheduling, not by reducing consumption.
- Efficiency ratio = rupees saved per kWh of total demand.

Graphical Insights (1/2)

Appliance Allocation

- Bar chart shows optimized energy equals required demand.
- Larger annual demand appliances receive larger optimized allocations.

Cost Comparison

- Side-by-side bars highlight the dramatic drop after optimization.
- The gap between bars is the monetary benefit of load shifting.

Graphical Insights (2/2)

Savings Contribution

- Pie/bar chart shows saved portion (1/3) of original bill.

Monthly Trend

- Optimized cost stays consistently below baseline every month.
- Savings are robust across different seasonal demand levels.

(All figures generated with `ggplot2` in R.)

Efficiency Ratio Interpretation

Efficiency ratio E is defined as:

$$E = \frac{S}{\sum_{i=1}^{n} R_i} = 3.38 \text{ Rs./kWh}$$

This means that for every kWh of total demand, shifting to off-peak saves Rs. 3.38. It provides a single intuitive measure of the economic value created per unit of energy handled under the optimized structure.

Model Strengths and Limitations

Strengths:

- Transparent formulation, easy to solve and interpret.
- Clearly demonstrates economic benefit of load shifting.
- Ideal for teaching optimization principles.

Limitations:

- No explicit hourly time indexing – assumes full shiftability.
- Ignores appliance-specific flexibility and capacity constraints.
- Stylized outcome, not a detailed operational schedule.

Possible Extensions

For a real-world smart-grid application:

- Introduce time-indexed variables x_{it} and hourly tariffs c_t.
- Add constraints for maximum shiftable load per appliance.
- Incorporate user preferences and usage patterns.
- Extend to mixed-integer programming for discrete on/off decisions.

Real-World Impact

- **Residential scale**: A 33% reduction can mean thousands of rupees saved per household annually.
- **Grid benefits**: Shifting load away from peak hours reduces strain on power infrastructure and helps avoid blackouts.
- **Environmental angle**: Lower peak demand may reduce reliance on expensive, polluting peaker plants.
- **Policy relevance**: Time-of-use tariffs incentivize such shifts; optimization models help consumers respond optimally.

This simple LP is a stepping stone toward more sophisticated demand-side management systems.

Summary

- Linear programming effectively minimizes electricity cost under time-varying tariffs.
- Achieved a **33.3% cost saving** (Rs. 7.96 million) for the given dataset.
- Savings arise entirely from scheduling, not from demand reduction.
- The work showcases translation of a real problem into mathematical form and computational solution.
- With added constraints, the framework can evolve into a realistic smart-energy tool.

Thank you!

Questions?